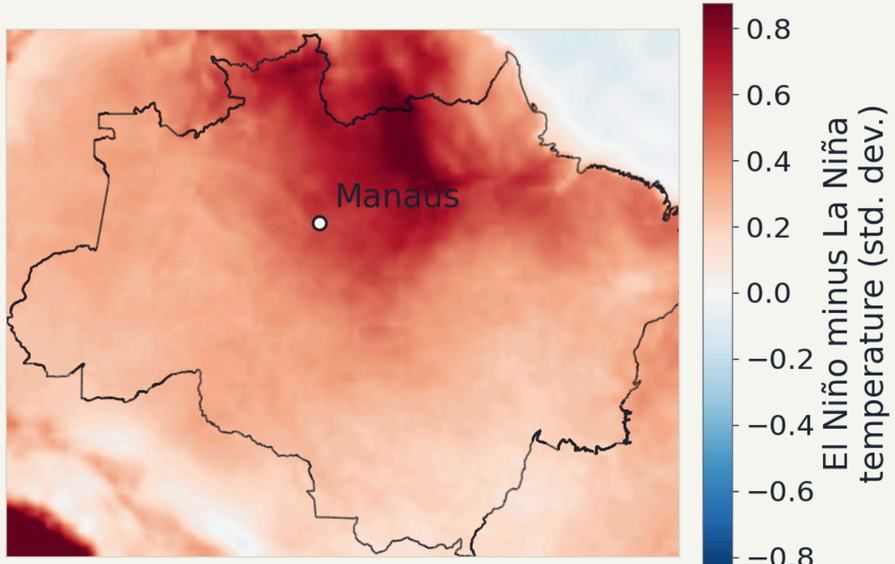
AMAZONIA · ERA5 · 1981–2025

Does El Niño leave a fingerprint on Amazonian temperature?

Self-supervised temperature regimes, tested against ENSO

Lívia Meinhardt · EMAp / FGV



Average temperature in El Niño months minus La Niña months

What we did

1

Daily temperature maps

Min and max temperature, relative to each place's normal for that month

2

AI groups similar days

Self-supervised model (Masked Siamese Network), no labels given

3

15 regimes

Every day is assigned to one recurring temperature pattern

4

Compare with ENSO

Which regimes become more or less common in El Niño or La Niña?

16,436
days analysed

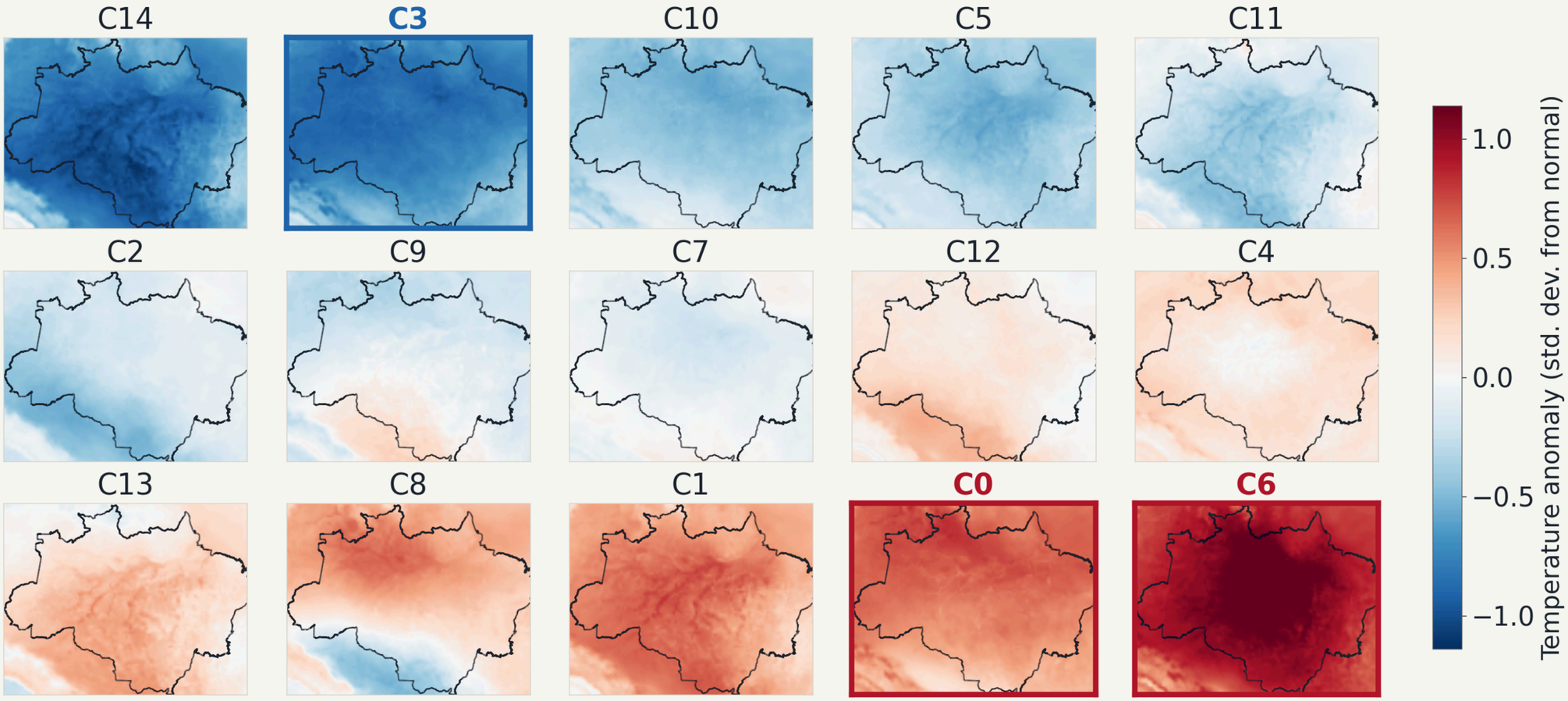
15
temperature regimes

35
ENSO episodes (11 El Niño)

Data: ERA5 reanalysis (Tmin, Tmax, 0.25°) · Oceanic Niño Index (NOAA)

The model found 15 temperature regimes

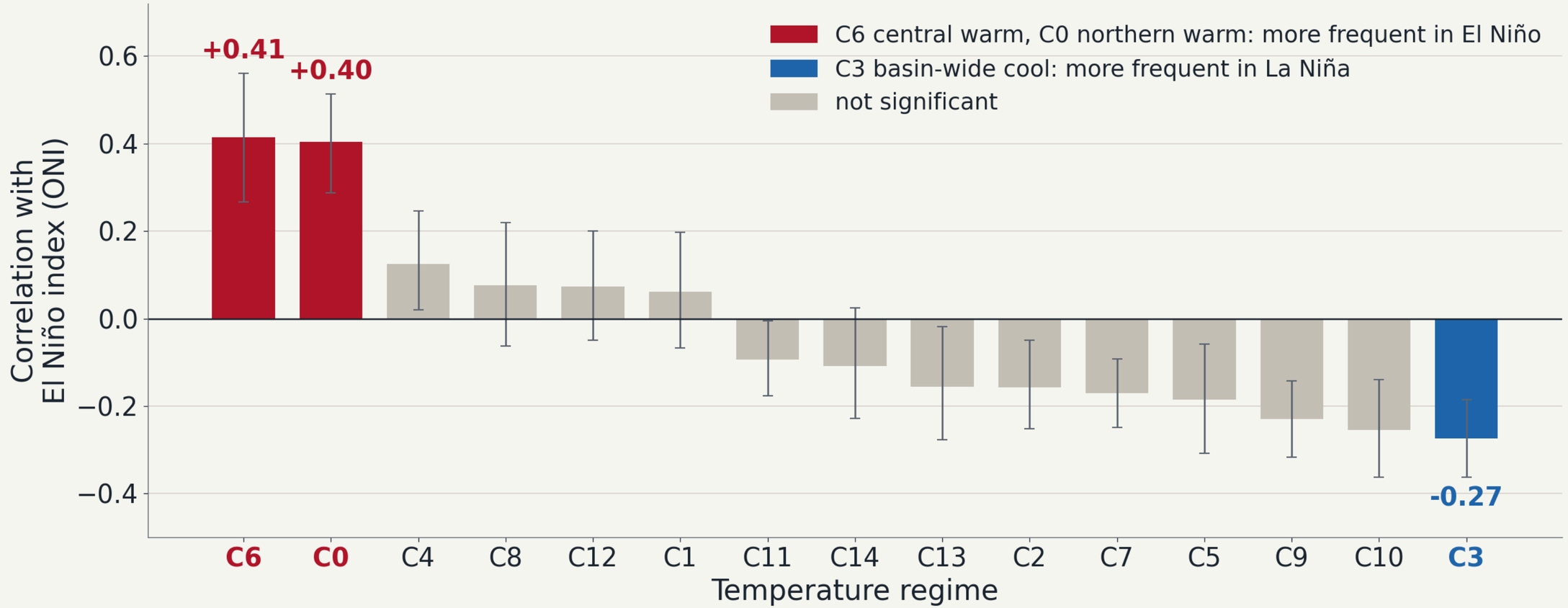
Average map of each regime, coldest to warmest. **Red** and **blue** frames: the 3 regimes linked to ENSO



Each regime covers 5.6–7.6% of days · black line: Legal Amazon border

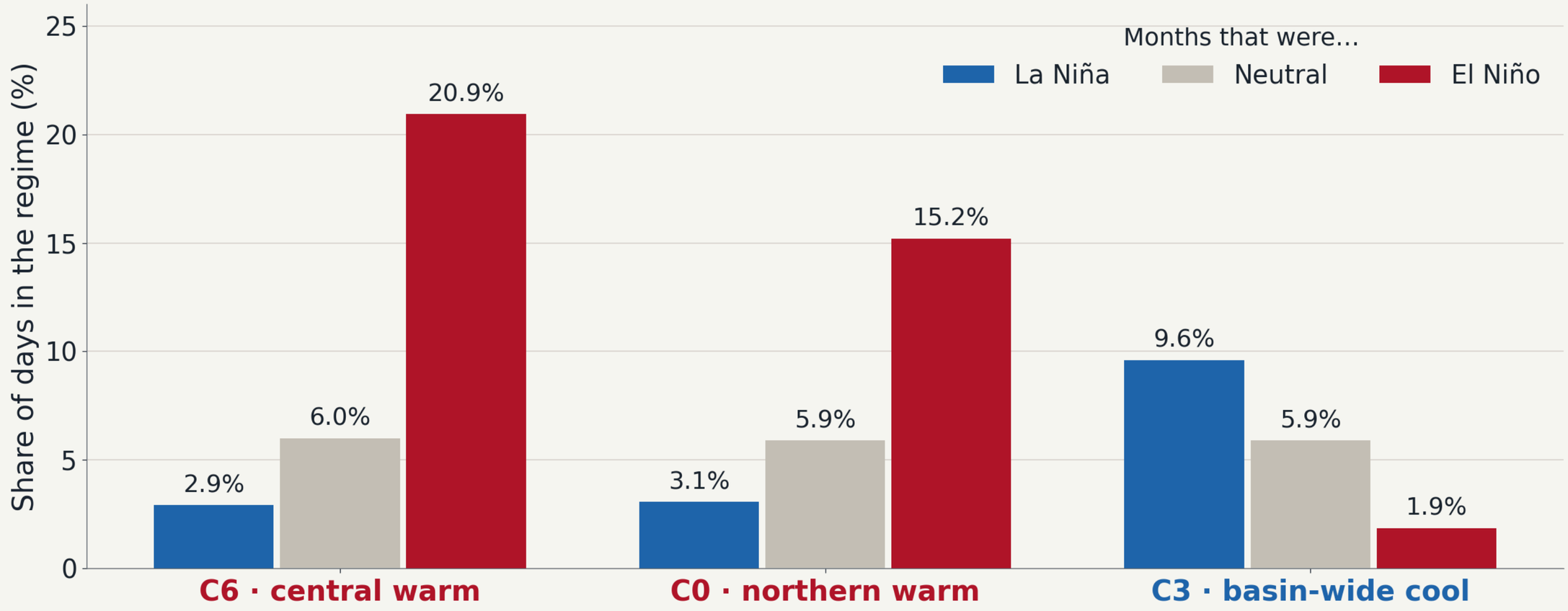
Three regimes follow El Niño

Two warm regimes rise with El Niño, one cool regime rises with La Niña



Monthly data, 1981–2025 · global test p = 0.013 · corrected for testing 15 regimes · bars: 95% CI

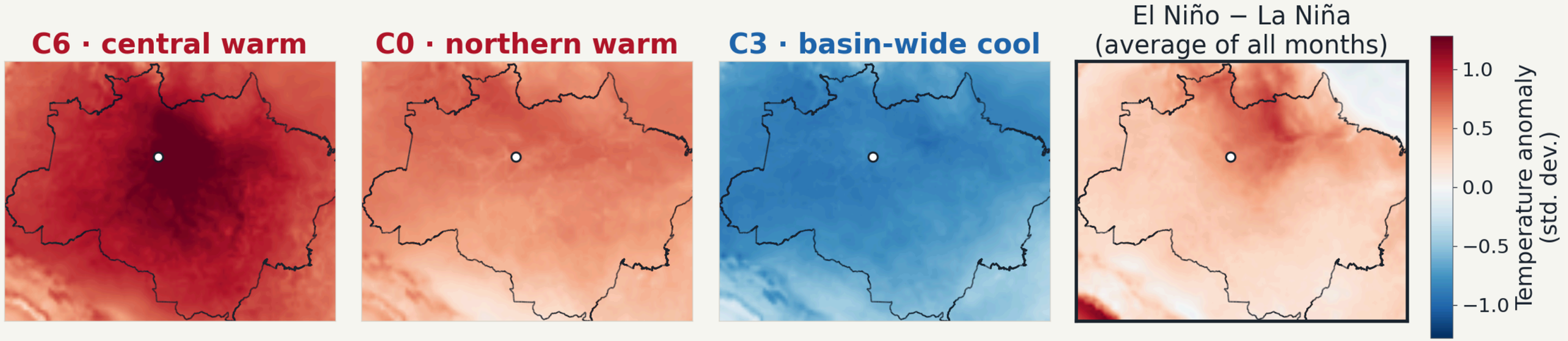
C6 is 7× more common in El Niño than in La Niña



531 months: 117 El Niño, 128 La Niña, 286 neutral (NOAA definition)

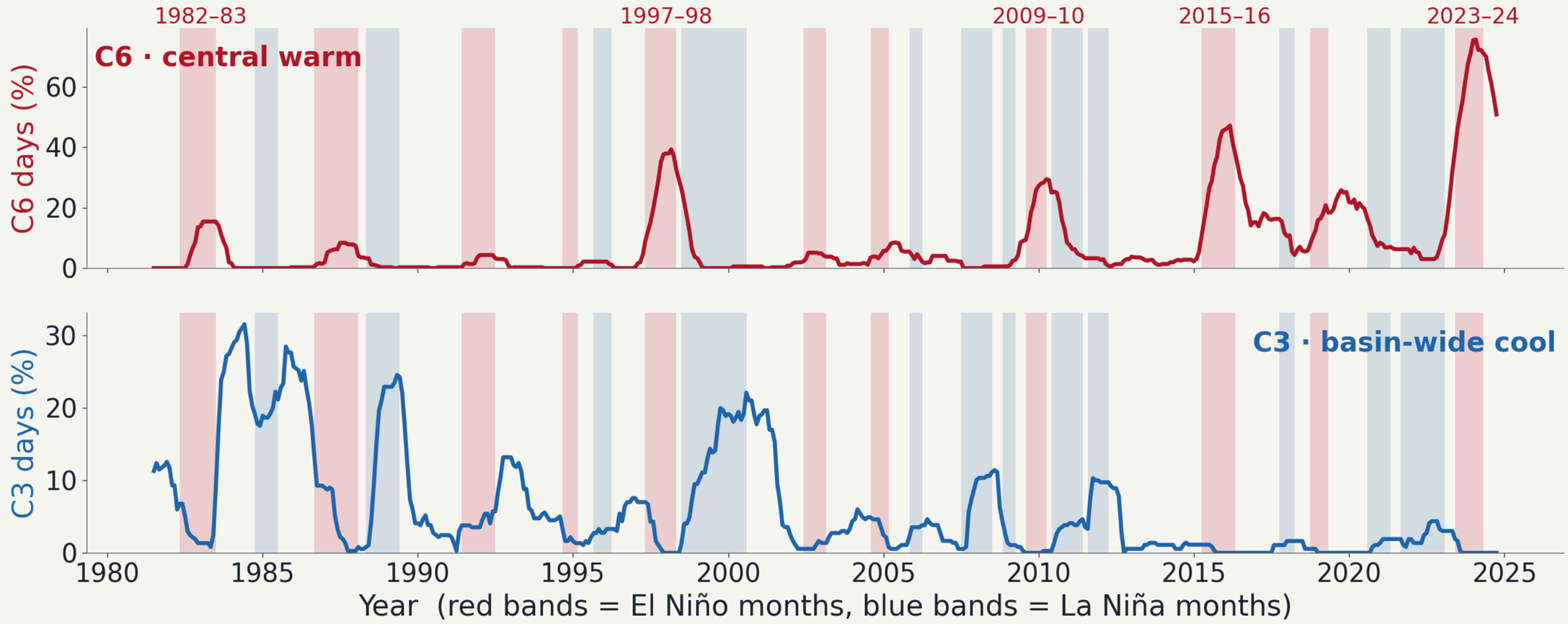
The ENSO regimes look like El Niño itself

The model never saw ENSO, yet its warm regimes match the El Niño pattern



Spatial correlation with the El Niño pattern: C6 +0.61 · C0 +0.67 · C3 −0.64

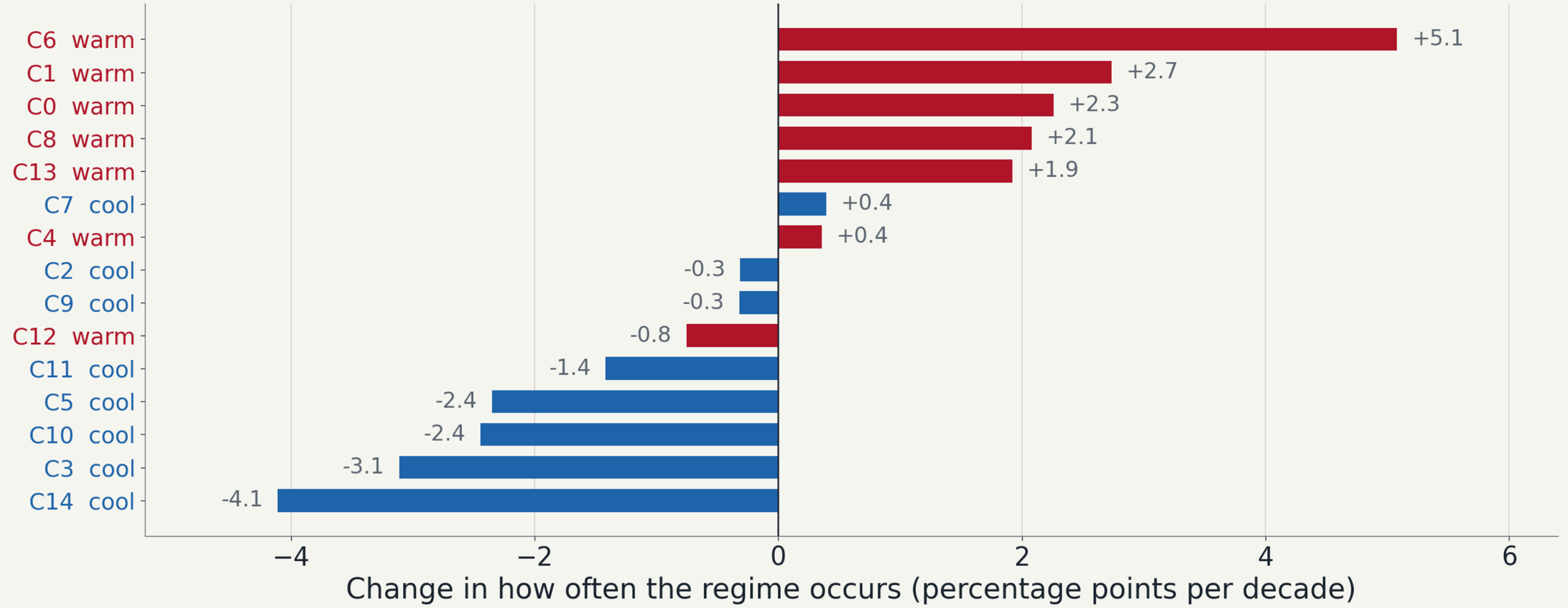
C6 peaks in every major El Niño



12-month rolling average of the share of days in each regime

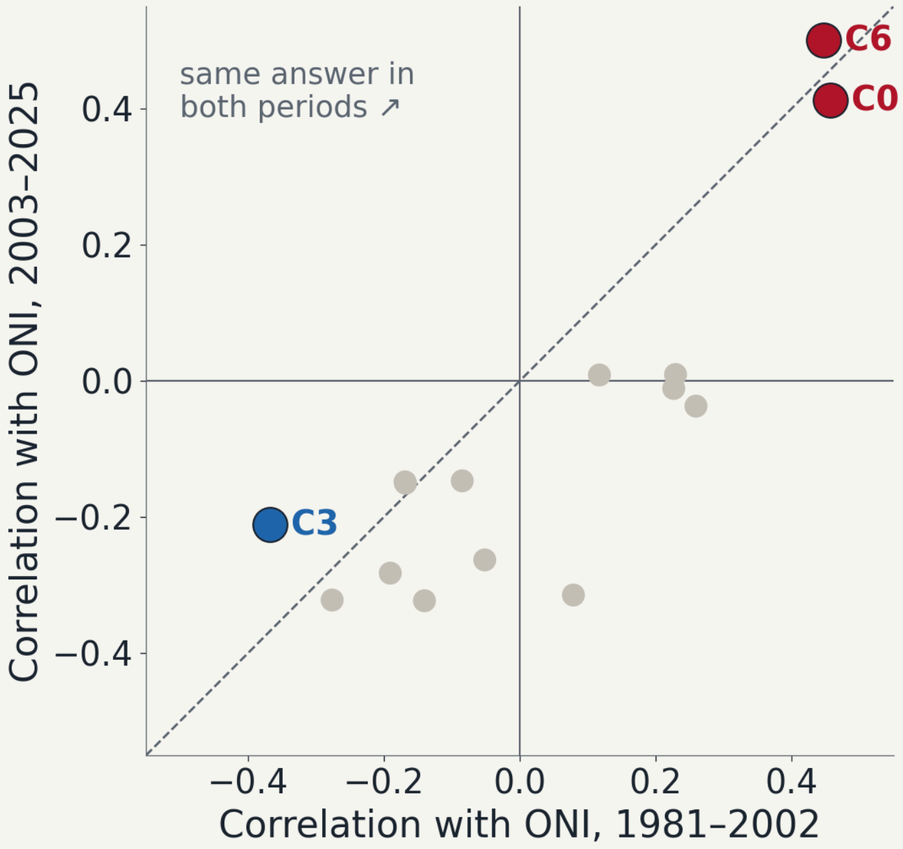
Warm regimes are becoming more common

The three fastest-growing regimes are warm; the three fastest-shrinking are cool



Linear trend of yearly frequency, 1981–2025

The result holds in both halves of the record



r = 0.83

Agreement between each regime's link to ENSO in **1981–2002** and in **2003–2025**, computed separately.

C6, C0 and C3 keep the same sign in both periods.

Each dot is one regime

Takeaways

15 An AI model with no labels finds meaningful, recurring temperature regimes in Amazonia.

3 Of them respond to ENSO robustly: two warm regimes with El Niño, one cool regime with La Niña.

+5.1 Percentage points per decade: the most El Niño-like regime (C6) is becoming more frequent.

Next: add precipitation, extend back to 1940